

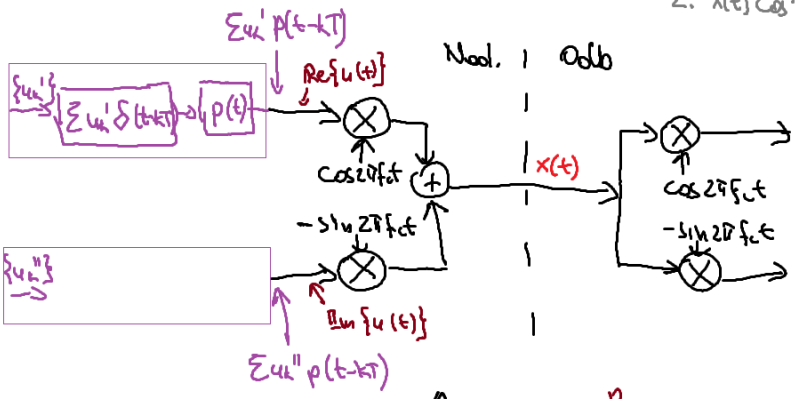
1. Dany jest sygnał $x(t)$

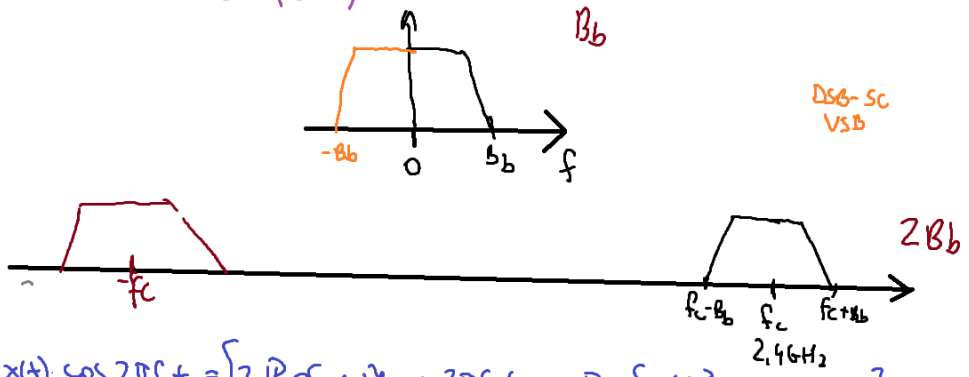
$x(t) = 2\,\mathrm{Re}\{u(t)\}\cos 2\pi f_c t - 2\,\mathrm{Im}\{u(t)\}\sin 2\pi f_c t$

Wynikiem sygnały odbierane i odbierany

$x(t)\cos 2\pi f_c t$ $\quad -x(t)\sin 2\pi f_c t$

1. Dlaczego tak, $x(t)$?
2. $x(t)\cos$ oraz $-x(t)\sin$
3. Pasmo





$$x(t)\cos 2\pi f_c t = \left[2\,\mathrm{Re}\{u(t)\}\cos 2\pi f_c t - 2\,\mathrm{Im}\{u(t)\}\sin 2\pi f_c t\right]\cos 2\pi f_c t =$$

$$= 2\,\mathrm{Re}\{u(t)\}\cos^2 2\pi f_c t - \mathrm{Im}\{u(t)\}\,2\sin 2\pi f_c t\cos 2\pi f_c t =$$

$2\cos^2\alpha = 1+\cos 2\alpha$

$$= \mathrm{Re}\{u(t)\}\left[1+\cos 4\pi f_c t\right] - \mathrm{Im}\{u(t)\}\sin 4\pi f_c t$$

$$= \mathrm{Re}\{u(t)\} + \mathrm{Re}\{u(t)\}\cos 2\cdot 2\pi f_c t - \mathrm{Im}\{u(t)\}\sin 2\cdot 2\pi f_c t$$

$x(t)(-\sin 2\pi f_c t)$ ← 2ga skł.

odfiltrowane

2. Wyznacz średnią energię dla QPSK oraz 16 QAM, (64-QAM) oraz dla M-QAM



$$\bar{E} = \frac{1}{M}\sum_{n=1}^{M} |d(n)|^2$$

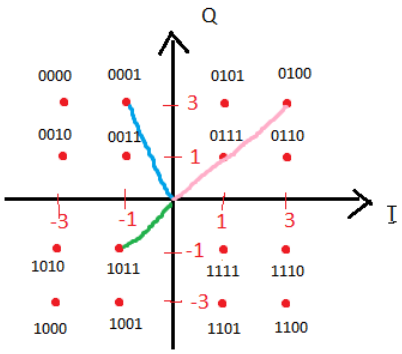
$$\bar{E}_{QPSK} = \frac{1}{4}\left(\left(\sqrt{2}\right)^2 + \left(\sqrt{2}\right)^2 + \left(\sqrt{2}\right)^2 + \left(\sqrt{2}\right)^2\right) =$$

$$= 2$$

3. Metryki dla symboli wg kodowania Greya

$\bar{E}_{16QAM} = 10$ $\qquad E_{16PAM} = 85$

$\bar{E}_{64QAM} =$ $\qquad E_{64PAM} =$



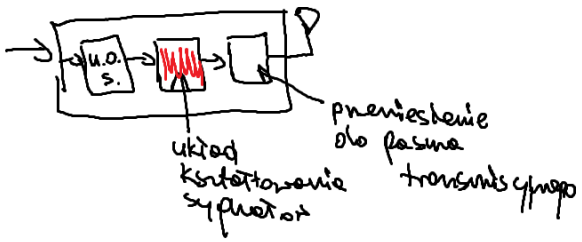
$$\sqrt{1^2+1^2} = \sqrt{2}$$

$$\sqrt{3^2+1^2} = \sqrt{10}$$

$$\sqrt{3^2+3^2} = \sqrt{18}$$

$$\overline{E}_{16QAM} = \frac{1}{16}\left[4\cdot\sqrt{2}^2 + 8\cdot\sqrt{10}^2 + 4\sqrt{18}^2\right] =$$

$$= \frac{1}{16}\left(8+80+72\right) = \underline{\underline{10}}$$

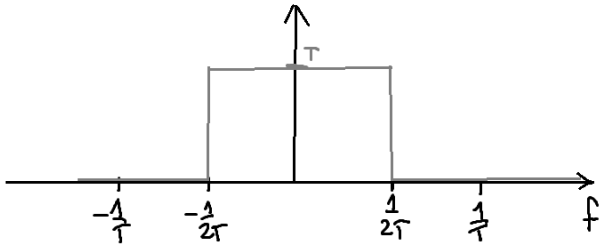


Dany jest impuls o kształcie podniesionego cosinusa:

$$G_\alpha(f) = \begin{cases} T & 0 \le |f| \le \frac{1-\alpha}{2T} \\ T\left(\cos\left[\frac{\pi T}{2\alpha}\left(|f| - \frac{1-\alpha}{2T}\right)\right]\right)^2 & \frac{1-\alpha}{2T} < |f| \le \frac{1+\alpha}{2T} \\ 0 & |f| > \frac{1+\alpha}{2T} \end{cases}$$

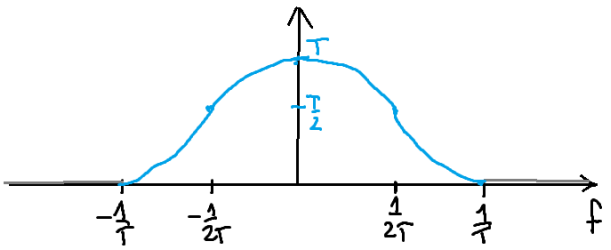
Narysuj impuls podn. cos. dla $\alpha = 0$, $\alpha = 1$, $\alpha = 0.15$, $\alpha = 0.25$

$\alpha = 0$



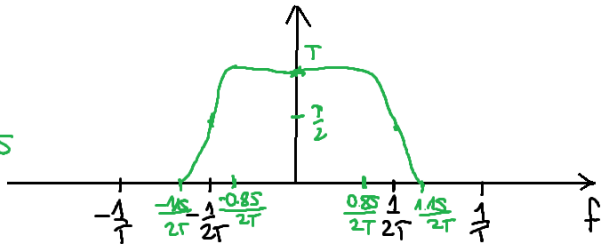
$\alpha = 1$

$\alpha = 0 \div 1$



$\alpha = 0.15$

$\alpha = 0.15 - 0.25$



$$T\left(\cos\left[\frac{\pi T}{2}\left(|f| - \frac{1-\alpha}{2T}\right)\right]\right)^2$$

$$T(\cos \pi \tfrac{T}{2} f)^2$$

$\frac{1}{2T}$ $T\left(\cos\frac{\pi}{4}\right)^2 =$

$T \cdot \left(\frac{\sqrt{2}}{2}\right)^2 = \frac{1}{2}T$

$\frac{1}{T}$ $T \cos\frac{\pi}{2} \to 0$

$$g_\alpha(t) = \mathrm{Sinc}\,\frac{t}{T}\,\frac{\cos(\pi\alpha t/T)}{1 - 4\alpha^2 t^2/T^2}$$

Wyznacz postać czasową impulsu na podstawie postaci częstotliwościowej dla $T = 1$ oraz $\alpha = 1$.

Przyjmij, że $g_1(f) = \frac{1}{2}\left[1 + \cos\left(\frac{\pi f}{2}\right)\right] \mathrm{rect}\frac{f}{2}$

Zastosuj wzór Eulera $\cos x = \frac{1}{2}\left[e^{ix} + e^{-ix}\right]$

Wskazówka: jako krok pośredni wyznacz: $g_1(t) = sinc\,(2t) + \frac{1}{2}sinc(2t+1) + \frac{1}{2}sinc(2t-1)$